Corrections to "A Signature-Constrained Two-Stage Framework for Hyperspectral Target Detection Based on Generative Self-Supervised Learning"

Zhengjun Cao and Olivier Markowitch

***Abstract*—The signature-constrained two-stage framework for hyperspectral target detection (HTD) (IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (IEEE TGRS), 5514917, 2025) needs to predict whether the label of each encoded feature vector belongs to the target category. The process computes the inner product between the feature vector and each signature feature vector. It claims that the product reflects the similarity between the two vectors, with higher values indicating greater similarity. We find this claim is generally not valid without further assumptions. We stress that the general inner product cannot be used to indicate the Euclidean distance between two vectors. We show that the distance-compatible similarity score can reflect the true similarity between any two vectors with the same dimension.**

***Index Terms*—Distance-compatible similarity score, hyperspectral target detection (HTD), signature feature vector, spectral–spatial similarity.**

I. INTRODUCTION

Recently, in the above article Qin et al. [1] have presented a signature-constrained two-stage framework for hyperspectral target detection (HTD) based on generative self-supervised learning. The framework has two processing stages: the pretraining stage, fine-tuning, and the detection stage. The first stage has three components: spectral–spatial masking, an autoencoder for spectral–spatial reconstruction, and a loss function based on spectral–spatial similarity. The second stage has three components: sample mining and tiling, fine-tuning, and inference fusion.

The spectral–spatial similarity loss (SSSL) is used to improve the reconstruction effect. It predicts whether the label of each encoded feature vector belongs to the target category by computing the inner product between the feature vector $\boldsymbol{F}$ and each signature feature vector $\boldsymbol{d}'$.

In this note, we show that the inner product of two vectors reflecting the similarity with higher values indicating greater similarity, is generally not valid without further assumptions (e.g., normalization). We would like to stress that the general inner product fails to indicate the Euclidean distance between two vectors. To indicate the true similarity between any two vectors with the same dimension, one needs to compute the distance-compatible similarity score so as to determine which vector in the set of target vectors is more similar to the querying vector. This finding is particularly helpful for researchers unfamiliar with the similarity scores of vectors.

II. SIMILARITY SCORES OF VECTORS

A. General Inner Product Fails to Indicate the Similarity

Given vectors $\boldsymbol{V}_1, \boldsymbol{V}_2, \boldsymbol{V}$ with the same dimension, to determine which is more similar to $\boldsymbol{V}$, it is usual to compute

$$d(\boldsymbol{V}_i, \boldsymbol{V}) = \|\boldsymbol{V}_i - \boldsymbol{V}\| = \sqrt{\|\boldsymbol{V}_i\|^2 - 2\boldsymbol{V}_i \cdot \boldsymbol{V} + \|\boldsymbol{V}\|^2} \tag{1}$$

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Zhengjun Cao is with the Department of Mathematics, Shanghai University, Shanghai 200444, China (e-mail: caozhj@shu.edu.cn).

Olivier Markowitch is with the Computer Sciences Department, UniversitéLibre de Bruxelles, 1050 Brussels, Belgium (e-mail: olivier.markowitch@ulb.ac.be).

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$i = 1, 2$, and $\|\boldsymbol{V}\|$ is the Euclidean norm of $\boldsymbol{V}$. We assert that $\boldsymbol{V}_1$ is more similar to $\boldsymbol{V}$, if $d(\boldsymbol{V}_1, \boldsymbol{V}) < d(\boldsymbol{V}_2, \boldsymbol{V})$.

The inner product of two vectors fails to represent their similarity. For example

$$\boldsymbol{X} = \{1, 0, 1, 1, 0\},\ \boldsymbol{Y} = \{1, 1, 1, 1, 0\},\ \boldsymbol{Z} = \{0, 0, 1, 1, 0\}.$$

It is easy to find that $\boldsymbol{X} \cdot \boldsymbol{Z} = \boldsymbol{Y} \cdot \boldsymbol{Z} = 2$, but

$$d(\boldsymbol{X}, \boldsymbol{Z}) = 1,\ d(\boldsymbol{Y}, \boldsymbol{Z}) = \sqrt{2}.$$

So, $\boldsymbol{X}$ is more similar to $\boldsymbol{Z}$, though they have the same inner product with the vector $\boldsymbol{Z}$.

B. Distance-Compatible Similarity Score

In 2009, Wong et al. [2] introduced the technique of scalar-product-preserving encryption, in which the distance-compatible similarity score [3] can be used to indicate the similarity between any two vectors. Its basic idea can be explained as follows.

1) *Prolonged Target Vector:* Given a row vector $\boldsymbol{p}$, set
$$\boldsymbol{p}' = \left(\boldsymbol{p}, -0.5\|\boldsymbol{p}\|^2\right).$$
2) *Padded Querying Vector:* For a querying row vector $\boldsymbol{q}$, set
$$\boldsymbol{q}' = (\boldsymbol{q}, 1).$$
3) *Similarity Score:* Compute the inner product
$$s = \boldsymbol{p}' \cdot \boldsymbol{q}'.$$

Theorem 1: Given two row vectors $\boldsymbol{p}$ and $\boldsymbol{q}$ with the same dimension, the similarity score

$$s = \left(\boldsymbol{p}, -0.5\|\boldsymbol{p}\|^2\right) \cdot (\boldsymbol{q}, 1). \tag{2}$$

It is compatible with the distance comparison routine.

Proof: Let $\boldsymbol{p}_1'$ and $\boldsymbol{p}_2'$ be two prolonged target vectors as above. Hence, we have

$$\begin{aligned}
&\left(\boldsymbol{p}_1' - \boldsymbol{p}_2'\right) \cdot \boldsymbol{q}' = \left(\boldsymbol{p}_1' - \boldsymbol{p}_2'\right) \boldsymbol{q}'^{\top} \\
&= \left(\left(\boldsymbol{p}_1, -0.5\|\boldsymbol{p}_1\|^2\right) - \left(\boldsymbol{p}_2, -0.5\|\boldsymbol{p}_2\|^2\right)\right)(\boldsymbol{q}, 1)^{\top} \\
&= \left(\boldsymbol{p}_1 - \boldsymbol{p}_2, -0.5\|\boldsymbol{p}_1\|^2 + 0.5\|\boldsymbol{p}_2\|^2\right)(\boldsymbol{q}, 1)^{\top} \\
&= \frac{1}{2}\left(2\boldsymbol{p}_1\boldsymbol{q}^{\top} - 2\boldsymbol{p}_2\boldsymbol{q}^{\top} - \|\boldsymbol{p}_1\|^2 + \|\boldsymbol{p}_2\|^2\right) \\
&= \frac{1}{2}\left(\|\boldsymbol{p}_2\|^2 - 2\boldsymbol{p}_2\boldsymbol{q}^{\top} + \|\boldsymbol{q}\|^2\right) - \frac{1}{2}\left(\|\boldsymbol{q}\|^2 - 2\boldsymbol{p}_1\boldsymbol{q}^{T} + \|\boldsymbol{p}_1\|^2\right) \\
&= \frac{1}{2}\left(\|\boldsymbol{p}_2 - \boldsymbol{q}\|^2 - \|\boldsymbol{p}_1 - \boldsymbol{q}\|^2\right) \\
&= \frac{1}{2}\left(d\left(\boldsymbol{p}_2, \boldsymbol{q}\right) + d\left(\boldsymbol{p}_1, \boldsymbol{q}\right)\right)\left(d\left(\boldsymbol{p}_2, \boldsymbol{q}\right) - d\left(\boldsymbol{p}_1, \boldsymbol{q}\right)\right).
\end{aligned}$$

Let $s_i = \boldsymbol{p}_i' \cdot \boldsymbol{q}'$, $i = 1, 2$. Since $d(\boldsymbol{p}_2, \boldsymbol{q}) + d(\boldsymbol{p}_1, \boldsymbol{q}) > 0$

$$\left(\boldsymbol{p}_1' - \boldsymbol{p}_2'\right) \cdot \boldsymbol{q}' > 0 \Leftrightarrow d\left(\boldsymbol{p}_2, \boldsymbol{q}\right) - d\left(\boldsymbol{p}_1, \boldsymbol{q}\right) > 0$$
$$s_1 > s_2 \Leftrightarrow d\left(\boldsymbol{p}_2, \boldsymbol{q}\right) > d\left(\boldsymbol{p}_1, \boldsymbol{q}\right).$$

Thus, the similarity score indicates the Euclidean distance between the target vector and the querying vector, with a higher score indicating greater similarity. ■

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III. Review of the Signature-Constrained Fine-Tuning

Given a hyperspectral image $\boldsymbol{X}$, split it into

$$\boldsymbol{S} = \text{Slice}\,(\boldsymbol{X}) = [\boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_i, \ldots]$$

where $\boldsymbol{S}_i \in \mathbb{R}^{P\times P\times I}$. For the patch $\boldsymbol{S}_i$, transform it into $P\times P$ embedded vectors. Then flatten and feed them to produce the feature token $\boldsymbol{F}$.

A signature $\boldsymbol{d}$ is projected into the D-dimensional embedding space to generate a feature vector $\boldsymbol{d}'$. This signature feature vector is then multiplied by each encoded feature vector to acquire the primary prediction map. A sigmoid function is subsequently applied to yield the predicted map R_i of the input hyperspectral image patch

$$R_i = \boldsymbol{d}' * \boldsymbol{F} \tag{3}$$

where $*$ represents the inner product operation. It stresses that "*the product reflects the similarity between the two vectors, with higher values indicating greater similarity*." See [1, eq. (17)] and its descriptions.

IV. Revised Similarity Score

As discussed above (see Section II), the general inner product R_i does not indicate the similarity between the feature vector $\boldsymbol{d}'$ and the target feature vector $\boldsymbol{F}_i$. To revise, one needs to use a distance-compatible similarity score

$$\hat{R}_i = (\boldsymbol{d}', 1) * \left(\boldsymbol{F}_i, -0.5\|\boldsymbol{F}_i\|^2\right). \tag{3'}$$

If the score $\hat{R}_\ell$ is the biggest, we then assert that the target feature vector $\boldsymbol{F}_\ell$ is the nearest to vector $\boldsymbol{d}'$.

Notice that if all vectors are normalized, the scalar product is strictly monotonic with Euclidean distance, because

$$d\,(\boldsymbol{V}_i, \boldsymbol{V}) = \sqrt{\|\boldsymbol{V}_i\|^2 - 2\boldsymbol{V}_i\cdot\boldsymbol{V} + \|\boldsymbol{V}\|^2} = \sqrt{2 - 2\boldsymbol{V}_i\cdot\boldsymbol{V}}$$
$$\boldsymbol{V}_i\cdot\boldsymbol{V} = 1 - \frac{1}{2}d\,(\boldsymbol{V}_i, \boldsymbol{V})^2.$$

But in the considered scenario, the feature vectors are not normalized so as to represent the hyperspectral targets. This note is essentially aimed at the "raw" geometric interpretation (without normalization or additional assumptions).

In the section of detection performance (see [1]), it claims

The area under the receiver operating characteristic curve (AUC) is utilized to evaluate the performance of the proposed framework. The AUC metric is considered a complete description of HTD detection performance. Specifically, $AUC_{(D,F)}$ and $AUC_{(F,\tau)}$ are important metrics for assessing detector performance, which reflect the effectiveness, the detection probability, and the BS capability of a detector, respectively. Generally, the higher $AUC_{(D,F)}$ and the smaller $AUC_{(F,\tau)}$ indicate better detector performance.

We remark the numerical values in [1, Tables I and II] should be re-evaluated experimentally with the new score, because they are calculated based on the general inner product without normalization. This note deals with the score as a purely geometric problem. It does not call into question the definition of SSSL, but only the interpretation of the score used for class prediction. Even if the Euclidean similarity interpretation is unsuitable, the revised scalar product can remain a valid architectural choice as a classification score, independently of the Euclidean distance. The correction would then better distinguish between a misinterpretation and an incorrect architectural choice.

V. Conclusion

We show that the general inner product of two vectors cannot be directly used to indicate their similarity unless vectors are normalized or additional structure is imposed. We stress that both vectors could be converted into new versions so as to calculate their distance-compatible similarity score. The findings in this note could be helpful for future work on this topic.

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References

[1] H. Qin, S. Wang, Y. Li, W. Xie, K. Jiang, and K. Cao, "A signature-constrained two-stage framework for hyperspectral target detection based on generative self-supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5514917.

[2] W. K. Wong, D. W.-L. Cheung, B. Kao, and N. Mamoulis, "Secure kNN computation on encrypted databases," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, Jun. 2009, pp. 139–152.

[3] Z. Cao, X. Wang, and L. Liu, "One privacy-preserving multi-keyword ranked search scheme revisited," *Int. J. Inf. Comput. Secur.*, vol. 16, nos. 3–4, pp. 375–384, 2021.

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